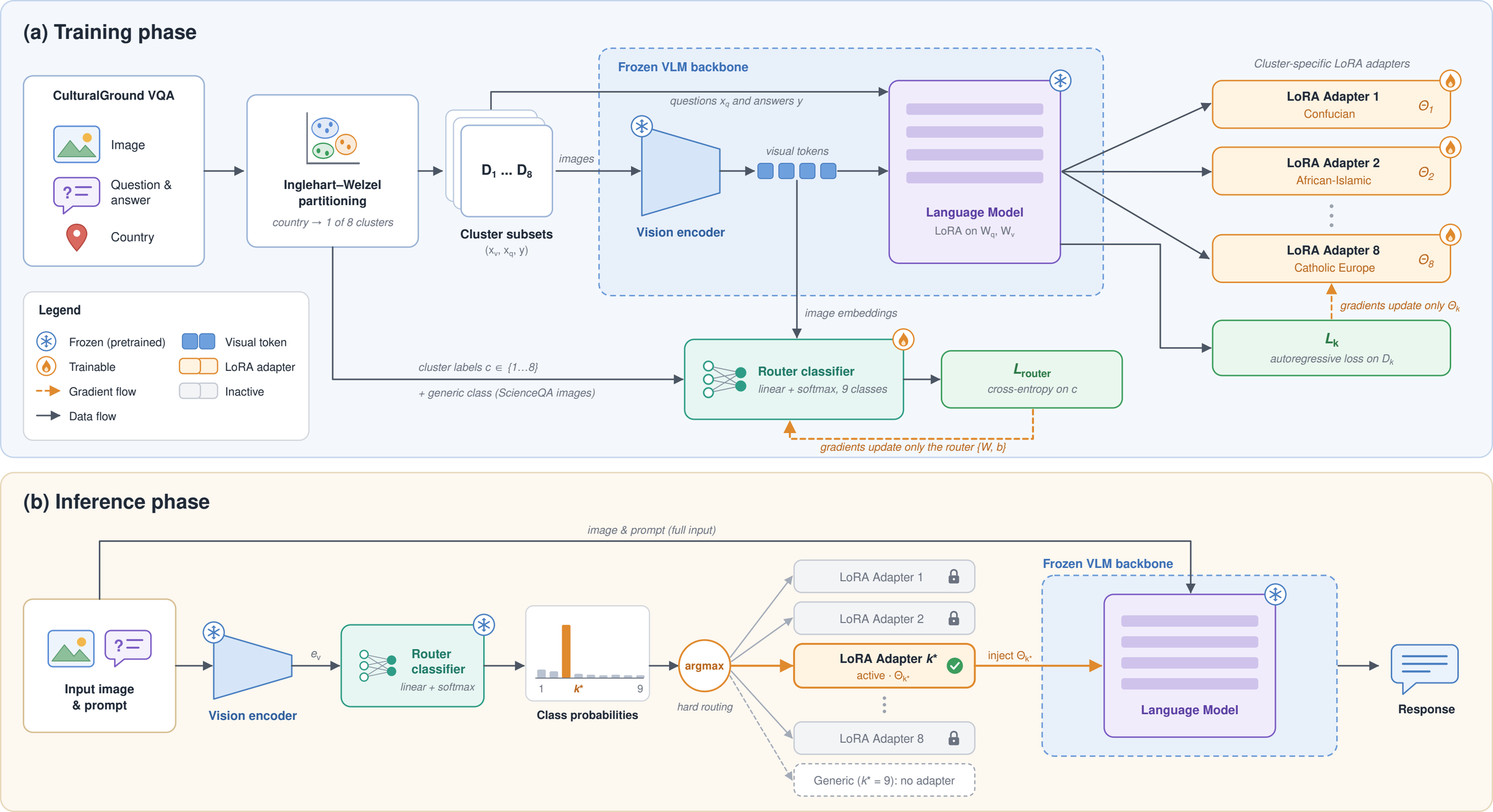

(a) Training phase
CulturalGround VQA
Image
Question & answer
Country
Inglehart–Welzel partitioning
country → 1 of 8 clusters
D1 ... D8
Cluster subsets
(xv, xq, y)
images
Frozen VLM backbone
questions xq and answers y
Vision encoder
visual tokens
Language Model
LoRA on Wq, Wv
Cluster-specific LoRA adapters
LoRA Adapter 1
Confucian
Θ1
LoRA Adapter 2
African-Islamic
Θ2
LoRA Adapter 8
Catholic Europe
Θ8
gradients update only Θk
Lk
autoregressive loss on Dk
image embeddings
cluster labels c ∈ {1…8}
+ generic class (ScienceQA images)
Router classifier
linear + softmax, 9 classes
Lrouter
cross-entropy on c
gradients update only the router {W, b}
Legend
Frozen (pretrained)
Trainable
Gradient flow
Data flow
Visual token
LoRA adapter
Inactive
(b) Inference phase
image & prompt (full input)
Input image & prompt
Vision encoder
ev
Router classifier
linear + softmax
1
k*
9
Class probabilities
argmax
hard routing
LoRA Adapter 1
LoRA Adapter 2
LoRA Adapter k*
active · Θk*
LoRA Adapter 8
Generic (k* = 9): no adapter
inject Θk*
Frozen VLM backbone
Language Model
Response